Hongyan Ji

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Education

The University of Iowa, Iowa City, IA

- Ph.D. in Computer Science 2021–present
  Expected completion: May 2027. Advisor: Sriram V. Pemmaraju.
- Graduate Certificate in College Teaching 2025
- B.S. in Computer Science and Master of Computer Science (M.C.S.) 2017–2021

Research Interests

Algorithms across computational models, especially distributed and parallel graph algorithms; approximation and randomization; communication, memory, and energy efficiency.

Publications

Conference papers

- **Approximating Minimum Dominating Set with Few Awake Rounds.** Hongyan Ji, Shreyas Pai, and Sriram V. Pemmaraju. 40th International Symposium on Distributed Computing (DISC), 2026. To appear.
  Invited to submit the full version to the DISC 2026 special issue of *Distributed Computing*.
- **Fast Deterministic Massively Parallel Ruling Sets Algorithms.** Hongyan Ji, Kishore Kothapalli, Sriram V. Pemmaraju, and Ajitanshu Singh. 26th International Conference on Distributed Computing and Networking (ICDCN), 2025.
- **Faster Set Cover in the MPC Model.** Hongyan Ji, Shreyas Pai, Sriram V. Pemmaraju, and Joshua Sobel. 26th International Conference on Distributed Computing and Networking (ICDCN), 2025. **Best Paper Award, Distributed Computing Track.**
- **Towards Singular Optimality in the Presence of Local Initial Knowledge.** Hongyan Ji and Sriram V. Pemmaraju. 31st International Colloquium on Structural Information and Communication Complexity (SIROCCO), pp. 300–317, 2024.

Journal articles

- **Faster Set Cover in the MPC Model.** Hongyan Ji, Shreyas Pai, Sriram V. Pemmaraju, and Joshua Sobel. *Theoretical Computer Science*, vol. 1086, Article 116238, 2026. Journal version of the ICDCN 2025 paper above.
- **Fast Deterministic Massively Parallel Ruling Sets Algorithms.** Hongyan Ji, Kishore Kothapalli, Sriram V. Pemmaraju, and Ajitanshu Singh. *Theoretical Computer Science*, accepted. Journal version of the ICDCN 2025 paper above.

Honors and Funding

- University of Iowa CS Department Travel Grant ($600), DISC 2026
- University of Iowa Graduate & Professional Student Government (GPSG) Travel Grant ($300) 2026
- **Best Paper Award**, Distributed Computing Track, ICDCN 2025
- ACM PODC Student Travel Grant ($150) 2025
- University of Iowa CS Department Travel Grant ($600), SIROCCO 2024
- NSF I-Corps Workshop, UI Innovators: workshop funding ($1,000) 2023

Teaching Experience

The University of Iowa

- **Sole Instructor**, CS:3210 Programming Languages and Tools: Go Spring 2026
  Designed and taught a Go course for 18 students; developed the syllabus, lectures, assignments, and assessments around a staged MapReduce project. Used written reflections, handwritten quizzes, and optional oral reassessment to assess students' reasoning and understanding.
- **Guest Instructor**, CS:4330 Theory of Computation Fall 2024
  Designed and delivered three lectures on context-free grammars through the Practicum in College Teaching.
- **Teaching Assistant**, CS:5620 Distributed Systems and Algorithms Fall 2024
  Graded assignments and held office hours. Designed and delivered two lectures on network decomposition through the Practicum in College Teaching.
- **Discussion Section Instructor**, CS:2210 Discrete Structures Spring 2023
  Led two discussion sections, graded homework, and held office hours.
- **Grader**, CS:3330 Algorithms (online section) Spring 2022
  Graded assignments and held office hours.
- **Grader**, CS:4640 Computer Security Spring 2021
  Graded assignments and held office hours.

Research Experience

- **Research Assistant**, The University of Iowa. Advisor: Sriram V. Pemmaraju.
  Fall 2021; Fall 2022; Fall 2023; Spring 2024; Spring 2025, Fall 2025 and Fall 2026.

Service and Outreach

Professional service

- Conference reviewer: DISC 2026; SOSA 2026; PODC 2025.

Departmental service, The University of Iowa

- Volunteer, CS faculty interviews Spring 2023 and 2024
- Organizer, Ph.D. Colloquium Fall 2022
- Board Member, Computer Science Graduate Student Board Spring 2022

Outreach and mentoring, The University of Iowa

- Panelist, graduate school panel for REU students Summer 2026
- Mentor, ExploreU STEM Summer Camp Summers 2021–2023
  Supported mathematical and computational activities for middle school girls.
- Assistant, International Programs – Life in Iowa 2018–2019

Talks and Presentations

- 40th International Symposium on Distributed Computing (DISC) Nov 2026 (scheduled)
  Approximating Minimum Dominating Set with Few Awake Rounds. Rome, Italy.
- Algorithms Reading Group, The University of Iowa Sep 2025
  Efficient Algorithms for Constructing Very Sparse Spanners and Emulators.
- 10th Iowa Computer Science Graduate Research Symposium Fall 2024
  The University of Iowa.
- 31st International Colloquium on Structural Information and Communication Complexity (SIROCCO) May 2024
  Towards Singular Optimality in the Presence of Local Initial Knowledge. Vietri sul Mare, Italy.
- Algorithms Reading Group, The University of Iowa Oct 2023
  Learning-Augmented Algorithms for Online Steiner Tree.